

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – FEB / MARCH 2025

Program : B.E. – Information Science and Engineering
Course Name : Theory of Computation
Course Code : IS54 (2021,2022 Batch)

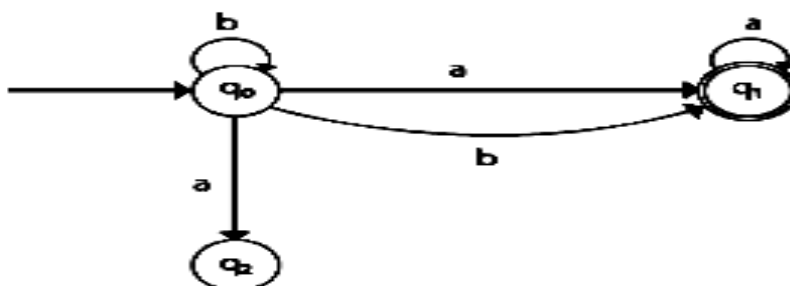
Semester : V
Max. Marks : 100
Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Construct a DFA to accepts strings of a's and b's having even numbers of a's and b's, and show that the input string aabbbb is valid. CO1 (10)
 - Is the following finite automaton DFA? Justify your answer. CO1 (04)



- What is the usage of ϵ -transitions in ϵ -NFA? Explain with an example. CO1 (06)
- Explain 5-tuple components $(Q, \Sigma, \delta, q_0, F)$ of DFA. CO1 (05)
 - Construct a NFA to accepts strings of a's and b's ending with 'ab', and show that the input string aabb is invalid. CO1 (05)
 - Convert the following ϵ -NFA to its equivalent DFA- CO1 (10)

δ	ϵ	a	b
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p, r\}$
q	$\emptyset$	$\{p\}$	$\emptyset$
*r	$\{p, q\}$	$\{r\}$	$\{p\}$

UNIT - II

- Illustrate the different operators of regular expression. Write regular expressions for the following languages: CO2 (10)
 - String of a's & b's containing not more than three a's.
 - $L = \{a^n b^m \mid n \geq 0, m \geq 0\}$
 - To accept a language consisting of string of a's and b's of even length.
 - State and prove that if $L = L(A)$ for some DFA A, then there is a regular expression R such that $L = L(R)$. CO2 (10)
- State and prove that if every language defined by a regular expression is also defined by a finite automaton. CO2 (10)
 - Construct the minimum state equivalent DFA: CO2 (10)

δ	a	B
$\rightarrow q_0$	q1	q3
q1	q2	q4
q2	q1	q4
q3	q2	q4
*q4	q4	q4

UNIT - III

5. a) Explain the following with examples:- CO3 (08)
 i) Nullable symbol ii) Unit-pair
 iii) Non-generating symbol iv) Non-reachable symbol.
- b) Define the following: CO3 (12)
 i. Parse Tree. ii. Ambiguous Grammar. iii. Leftmost Derivation.
 iv. Rightmost Derivation.
 Given Context Free Grammar $G = (\{S, C, a, b, e, t\}, \{a, b, e, t\}, R, S)$
 where $R = \{ S \rightarrow iCtS, S \rightarrow iCtSeS, S \rightarrow a, C \rightarrow b \}$. Obtain leftmost and
 rightmost derivation for the string $ibtibtibtaea$. Justify whether the
 given grammar G is ambiguous.
6. a) Design the context free grammar for the following languages: CO3 (06)
 i. $L = \{a^i b^j \mid i \leq 2j\}$
 ii. $L = \{a^n b^m c^m \mid n, m \geq 0\}$
 iii. $L = \{a^i b^j c^k \mid i = j + k\}$
- b) Convert the following CFG grammar $G = (V, \Sigma, R, S)$ to Chomsky CO3 (10)
 Normal Form, Where
 $R = \{$
 $S \rightarrow aTVa$
 $T \rightarrow aTa \mid bTb \mid \epsilon \mid V$
 $V \rightarrow cVc \mid \epsilon \}$
- c) List the applications of CFG. Explain any one of them. CO3 (04)

UNIT- IV

7. a) Discuss the graphical notation for PDA, with an example. CO4 (04)
 b) Define the language of PDA with empty stack and final state. Give CO4 (06)
 examples for each.
 c) Construct a PDA to accept the following language: wCw^R $w \in \{a,b\}^*$. CO4 (10)
 Show with the IDs that $abCba$ is valid.
8. a) Explain the working model of PDA, with a neat diagram. CO4 (10)
 b) Discuss how Deterministic PDA differs from Nondeterministic PDA. CO4 (06)
 c) Convert the grammar $G = (V, \Sigma, R, S)$ Where CO4 (04)
 $R = \{$
 $S \rightarrow 0AA$
 $A \rightarrow 0S \mid 1S \mid 0$
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 to a PDA that accepts the some languages by empty stack.

UNIT - V

9. a) Describe the halting problem in TM. CO5 (05)
 b) Write only transitions of the TM which performs subtraction CO5 (10)
 operation. Trace the transitions for the input string 0000100B.
 c) Explain NP-hard and NP-complete with an example. CO5 (05)
10. a) Explain the working principles of TM with a neat diagram. CO5 (08)
 b) Construct a TM to compute modulus operation. CO5 (12)
 i) Write the transition table.
 ii) Show the sequence of moves made for the string 0000100B.
